

Evidential Uncertainty-guided Mitochondria Segmentation for 3D EM Images

Motivation

- Background:**
- Recent progress in deep learning has enhanced the segmentation of mitochondria.
- Deep learning models are prone to producing over-confident yet inaccurate predictions.
- The utilization of uncertainty to rectify initial predictions would enhance the segmentation performance.

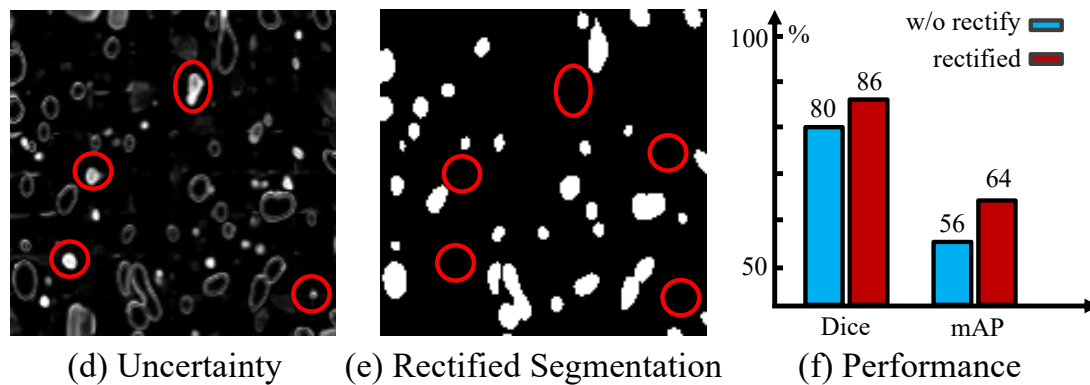
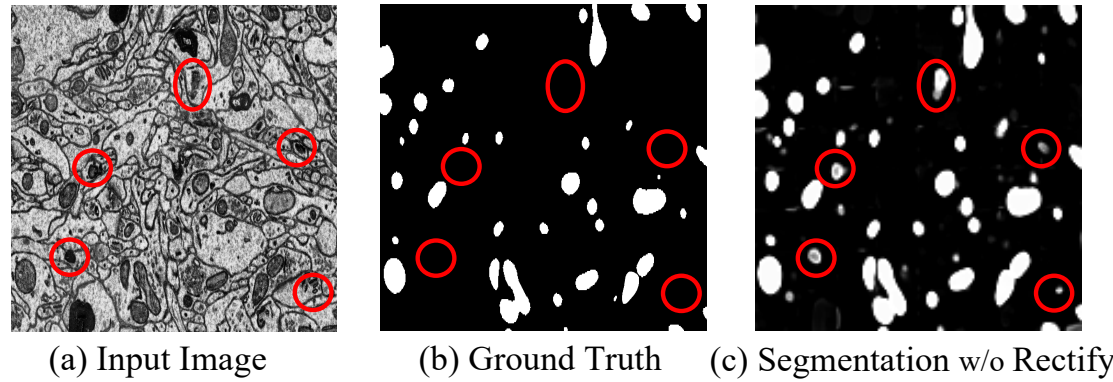
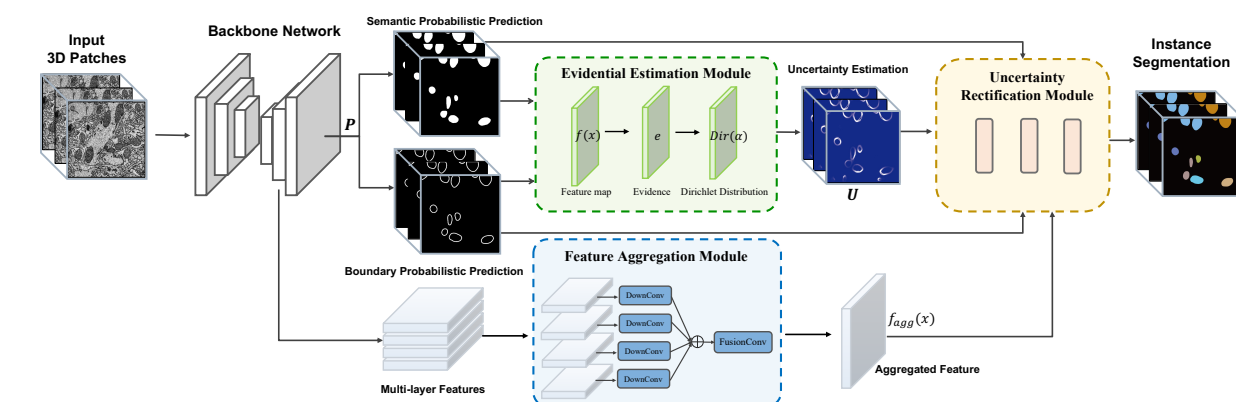


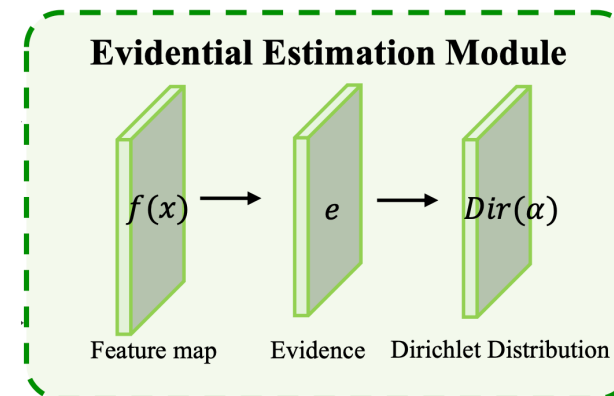
Figure 1. Illustration of the effectiveness of using uncertainty information to modify segmentation results.

Pipeline



Evidential Uncertainty Estimation

- EDL:** The evidential uncertainty estimation method (EDL) is a generalization of Bayesian theory to subjective probability. It assigns belief masses to each possible class label, and then the belief distribution of DST in the framework can be formalized as a Dirichlet distribution by Subjective Logic.



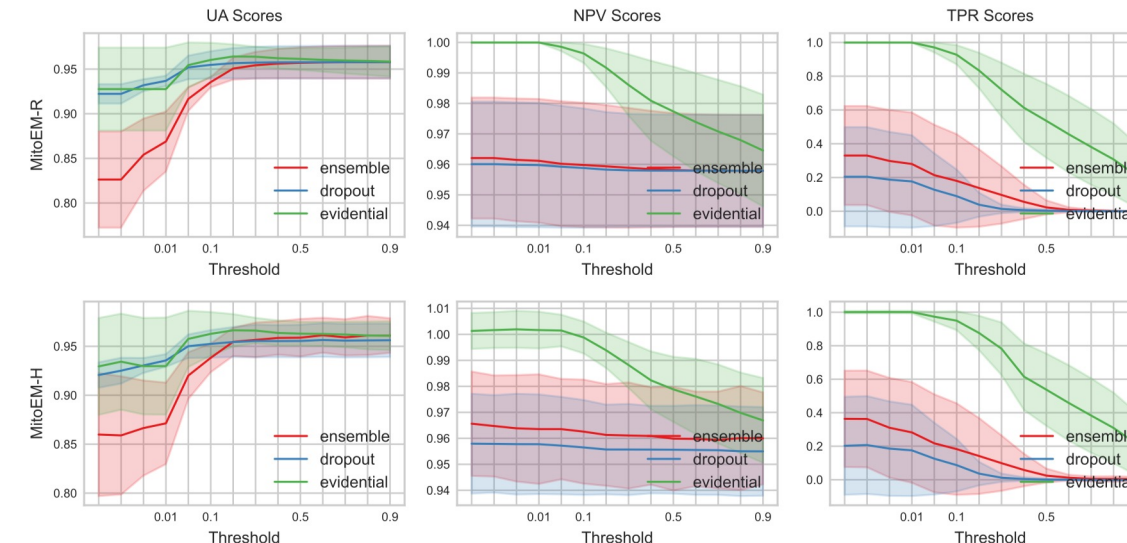
$$S = \sum_{k=1}^K \alpha_k$$

$$e_k = \alpha_k - 1$$

$$p_k = \alpha_k / S$$

$$u = \frac{K}{S}$$

- Calibration:** Evidential-based method could be more effective to rectify the erroneous predictions.



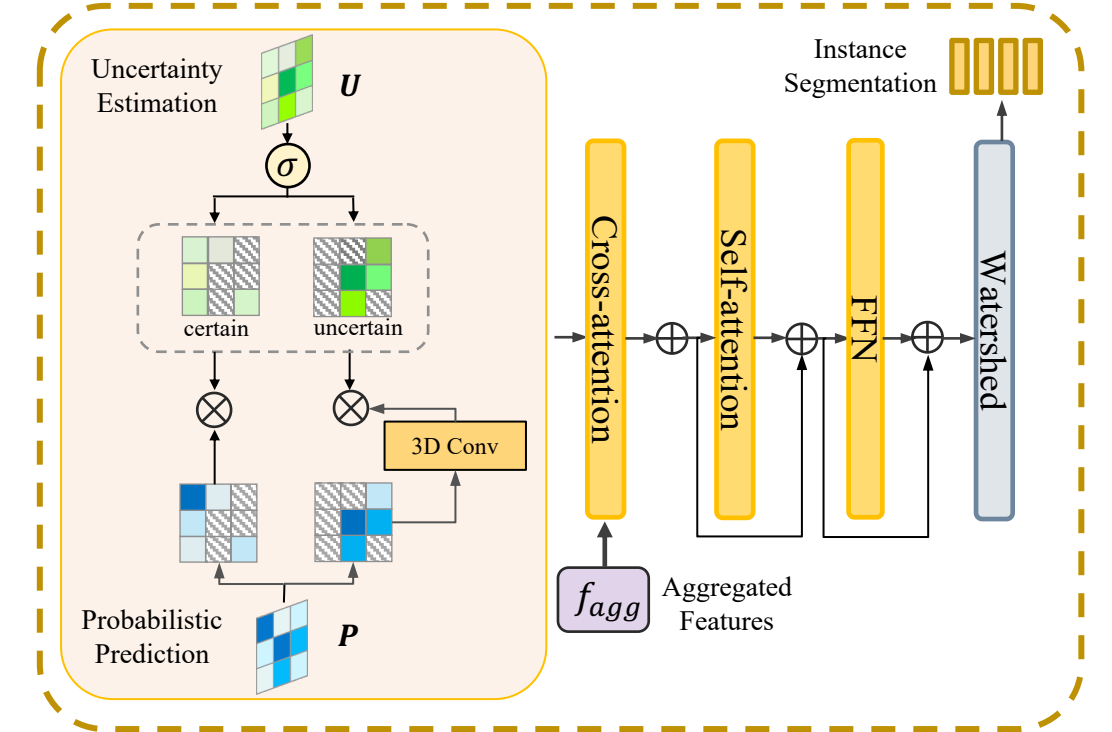
Loss Functions

$$1. \mathcal{L}_{EDL}^i = \sum_{k=1}^K y_{ik} (\psi(S_i) - \psi(\alpha_{ik})).$$

$$2. \mathcal{L}_{KL}^i = \log \left(\frac{\Gamma(\sum_{k=1}^K \tilde{\alpha}_{ik})}{\Gamma(K) \prod_{k=1}^K \Gamma(\tilde{\alpha}_{ik})} \right) + \sum_{k=1}^K (\tilde{\alpha}_{ik} - 1) [\psi(\tilde{\alpha}_{ik}) - \psi(\sum_{k=1}^K \tilde{\alpha}_{ik})].$$

$$3. \mathcal{L}_{Dice}^i = \sum_{k=1}^K \left(1 - \frac{2y_{ik}p_{ik}^{final}}{y_{ik} + p_{ik}^{final}} \right).$$

Rectify Segmentation Predictions

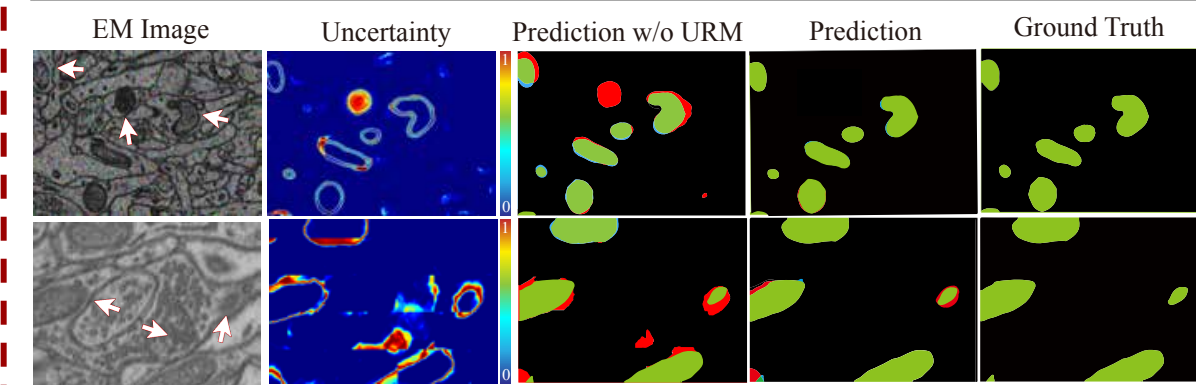


- Certain Case: $p_{ik}^{rec} = p_{ik} \cdot (1 - u_i)$
- Uncertain Case: $p_{ik}^{rec} = p_{ik}^{conv} \cdot (1 - u_i)$

Quantitative Results and Visualization

- Results:** The experimental results demonstrate the effectiveness to enhance 3D mitochondria segmentation on 4 datasets.

Methods	Dataset / Metric	MitoEM-H		MitoEM-R		Kasthuri++		Lucchi++	
		Dice	mAP	Dice	mAP	Dice	mAP	Dice	mAP
ConvNets-based	Xiao (Xiao et al. 2018)	0.798	0.812	0.830	0.903	0.947	0.900	0.882	0.910
	Peng and Yuan (Peng and Yuan 2019)	0.757	0.793	0.802	0.848	0.909	0.833	0.893	0.806
	U3D-BC (Wei et al. 2020)	0.746	0.773	0.775	0.844	0.889	0.831	0.880	0.753
	Zhili (Li et al. 2021)	0.765	0.787	0.794	0.870	0.935	0.909	0.865	0.811
	HIVE-Net (Yuan et al. 2021)	0.825	0.856	0.851	0.901	0.862	0.904	0.893	0.841
	nnU-Net (Isensee et al. 2021)	0.807	0.830	0.825	0.864	0.859	0.872	0.856	0.829
	Res-Unet3D (Li et al. 2022)	0.783	0.828	0.815	0.917	0.943	0.892	0.889	0.769
	nnFormer (Zhou et al. 2021)	0.787	0.830	0.824	0.813	0.919	0.874	0.889	0.825
Transformer-based	SwinUNETR (Hatamizadeh et al. 2022a)	0.779	0.822	0.803	0.867	0.904	0.861	0.869	0.874
	DSTUnet (Lin et al. 2022)	0.762	0.799	0.836	0.824	0.916	0.867	0.896	0.802
	TransBTS (Wang et al. 2021)	0.801	0.827	0.866	0.909	0.931	0.899	0.902	0.896
	UNETR (Hatamizadeh et al. 2022b)	0.775	0.809	0.813	0.836	0.855	0.858	0.872	0.801
	3D UX-Net (Lee et al. 2023)	0.816	0.845	0.859	0.901	0.950	0.917	0.902	0.910
	EUMS-3D (HIVE-Net)	0.837	0.871	0.865	0.911	0.895	0.909	0.907	0.869
EUMS-based	EUMS-3D (Res-Unet3D)	0.818	0.848	0.839	0.923	0.951	0.915	0.900	0.806
	EUMS-3D (SwinUNETR)	0.803	0.841	0.824	0.882	0.930	0.879	0.918	0.924
	EUMS-3D (TransBTS)	0.823	0.870	0.878	0.917	0.945	0.914	0.919	0.913
	EUMS-3D (3D UX-Net)	0.845	0.901	0.890	0.928	0.972	0.931	0.937	0.953



For predictions, green: true positive, red: false positive, blue: false negative.